

# SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

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ABSTRACT.

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## 1. INTRODUCTION

### 1.1. Background and motivation.

**Theorem 1.1.** *Let  $R$  be a normalised root system in  $\mathbb{R}^N$  and let  $G$  be the corresponding finite reflection group. Let  $k : R \rightarrow \mathbb{R}_+$  be  $G$ -invariant multiplicity function. Let  $R_{\text{Dunkl},j}$  be the  $j$ -th Dunkl-Riesz transform corresponding to  $R$  and  $k$ . Let  $w$  be the Dunkl weight corresponding to  $R$  and  $k$ . Let  $f \in L_1^{\text{loc}}(\mathbb{R}^N)$ .*

- (i) *If  $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$ ,  $p > N$ , then  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$ . Furthermore,*

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))} \leq c_{p,R,k} \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)};$$

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- (ii) If  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$ ,  $p > N$ , then  $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$ . Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))} \leq c_{p,R,k}^{-1} \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)};$$

- (iii) If  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ , then  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ . Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_{R,k} \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

- (iv) If  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ , then  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ . Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \geq c_{R,k}^{-1} \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

**1.2. Strategy of the proof.** In Section 3, we introduce our *ad hoc* Sobolev space and demonstrate that its semi-norm is controlled by the usual Sobolev semi-norm. In Section 4, we use Janson-Wolff method to control the  $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator  $[R_{\text{Dunkl},j}, M_f]$  by the *ad hoc* Sobolev semi-norm of  $f$ . Thus, we control the  $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator  $[R_{\text{Dunkl},j}, M_f]$  by the usual Sobolev semi-norm (i.e., we prove Theorem 1.1 (i)).

In Section 5, we control the oscillation semi-norm of  $f$  (with respect to measure  $w(x)dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$ ) via  $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator  $[R_{\text{Dunkl},j}, M_f]$ . After that, we verify the assumptions in Proposition 1.2 in [6] for measures  $w(x)dx$  and  $dx$  and, using the latter proposition, conclude that oscillation semi-norm of  $f$  with respect to measure  $w(x)dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$  is equivalent to the one with respect to measure  $dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$ . The latter is known to be equivalent to the usual Sobolev semi-norm of  $f$ .

**1.3. Notation.** Conventionally, we let  $\mathbb{N}$  denote the set of positive integers and set  $\mathbb{Z}_+ := \{0\} \cup \mathbb{N}$ . Throughout the paper, the symbols  $C$  and  $c$  denote positive constants (possibly varying from line to line) that may depend only on  $N$ ,  $p$ , and the structural data  $(R, k)$  of the Dunkl setting, but are independent of the main parameters (e.g., the symbol  $b$  and auxiliary decay factors). We write  $A \lesssim B$  to mean that  $A \leq CB$  for some constant  $C > 0$ , and  $A \sim B$  to mean that both  $A \lesssim B$  and  $B \lesssim A$  hold. For  $\lambda > 0$  and a cube  $Q$  with sidelength  $\ell(Q)$ , write  $\lambda Q$  for the cube concentric with  $Q$  and sidelength  $\lambda \ell(Q)$ . Similarly, for every ball  $B = B(x, r)$ , set  $\lambda B := B(x, \lambda r)$ . For any  $1 \leq p \leq \infty$ , we denote by  $p'$  the conjugate of  $p$ , which satisfies  $\frac{1}{p} + \frac{1}{p'} = 1$ . We denote the average of a function  $f$  over a  $\mu$ -measurable set  $E$  by

$$(1.1) \quad (f)_E := \int_E f(x) d\mu(x) := \frac{1}{\mu(E)} \int_E f(x) d\mu(x).$$

For a subset  $E \subset \mathbb{R}^N$ , we denote by  $\chi_E$  its characteristic function.

Denote by  $\{e_j\}_{j=1}^N$  the standard orthonormal basis on  $\mathbb{R}^N$ .

## 2. PRELIMINARIES

**2.1. Harmonic analysis in the Dunkl setting.** Let  $\mathbb{R}^N$  be the Euclidean space equipped with the usual scalar product  $\langle \cdot, \cdot \rangle$  and the induced norm  $\|\cdot\|$ . For a

nonzero vector  $v \in \mathbb{R}^N$ , the reflection  $\sigma_v$  with respect to the hyperplane orthogonal to a normalised vector  $v$  is defined by

$$\sigma_v x := x - 2\langle x, v \rangle v.$$

A finite set  $R \subset \mathbb{R}^N \setminus \{0\}$  is called a root system if  $\sigma_v(R) = R$  for every  $v \in R$ . A root system  $R$  is called a normalized reduced root system if  $|v| = 1$  for every  $v \in R$ . The finite group  $G$  generated by the reflections  $\{\sigma_v\}_{v \in R}$  is called the Weyl group (reflection group) of the root system  $R$ . Let  $k : R \rightarrow [0, \infty)$  be a multiplicity function, which is a  $G$ -invariant function fixed throughout the paper.

Define the Dunkl measure  $w(x)dx$  on  $\mathbb{R}^N$  by

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{k(v)}.$$

Let  $B(x, r) := \{y \in \mathbb{R}^N : |x - y| < r\}$  be the Euclidean ball centred at  $x$  with radius  $r > 0$ . Furthermore,

$$(2.1) \quad w(B(x, r)) \sim r^N \prod_{v \in R} (|\langle x, v \rangle| + r)^{k(v)},$$

where the implicit constant only depends on  $N, R$  and  $k$ . It follows from (2.1) that there exists a constant  $C \geq 1$  such that for every  $x \in \mathbb{R}^N$ ,

$$(2.2) \quad C^{-1} \left( \frac{r_2}{r_1} \right)^N \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left( \frac{r_2}{r_1} \right)^{N'}, \text{ for } 0 < r_1 < r_2,$$

so the numbers  $N'$  and  $N$  are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators  $T_\xi$  are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{k(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where  $\partial_\xi f$  denotes the directional derivative of  $f$  in the direction  $\xi$ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} = - \sum_{j=1}^N T_{e_j}^2.$$

In this article, we consider the Dunkl-Riesz transforms defined by  $R_{\text{Dunkl}, j} := T_{e_j} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}$ ,  $j = 1, 2, \dots, N$ .

**Definition 2.1.** The Weyl chambers are defined as the closures of connected components of the set

$$(2.3) \quad \{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}.$$

Define

$$d(x, y) = \min_{\sigma \in G} |x - \sigma(y)|, \quad x, y \in \mathbb{R}^N.$$

Recall from [5, Chapter VII, proof of Theorem 2.12] that

$$(2.4) \quad d(x, y) = |x - y| \text{ if and only if } x, y \in \Gamma \text{ for some chamber } \Gamma.$$

Denote by  $K_{\text{Dunkl}, j}(x, y)$  the integral kernel of  $R_{\text{Dunkl}, j}$ . We recall the kernel estimates for the Riesz transforms in the Dunkl setting proved in [4]. Although

these estimates may not be sharp (see the comment following Theorem 2.1 in [4, p. 10]), they suffice for our purposes.

**Lemma 2.2.** *There exists a constant  $C$  such that for  $j \in \{1, 2, \dots, N\}$  and every  $x, y$  with  $d(x, y) \neq 0$ ,*

$$|K_{\text{Dunkl},j}(x, y)| \leq \frac{d(x, y)}{|x - y|} \cdot \frac{C}{w(B(x, d(x, y)))}.$$

*Proof.* The proof is given in [4, Theorem 1.1].  $\square$

**Lemma 2.3.** *We have*

$$d(x, y)^2 \cdot w(x)w(y) \leq Cw^2(B(x, d(x, y))), \quad x, y \in \mathbb{R}^N.$$

*Proof.* Define equivalence relation  $\sim$  on  $R$  by saying  $v_1 \sim v_2$  if there exists  $g \in G$  with  $v_1 = gv_2$ . Let us split  $R$  into equivalence classes with respect to  $\sim$ . Since  $k$  is  $G$ -invariant, it follows that  $k$  is constant on each equivalence class. Note that  $G$  leaves each equivalence class invariant.

We, therefore, write

$$w(x) = \prod_{A \in R/\sim} \left( \prod_{v \in A} |\langle x, v \rangle| \right)^{k|A|}.$$

On the other hand,

$$w(B(x, r)) \geq c_{R,k}^{(1)} r^N \prod_{A \in R/\sim} \left( \prod_{v \in A} (r + |\langle x, v \rangle|) \right)^{k|A|}.$$

Let  $x \in \Gamma$  and  $y \in g\Gamma$ . Set  $z = g^{-1}y \in \Gamma$ . Note that  $d(x, y) = |x - z|$  and, therefore,

$$|\langle x, v \rangle| + d(x, y) = |\langle x, v \rangle| + |x - z| \geq 2^{-\frac{1}{2}} |\langle z, v \rangle|.$$

Thus,

$$\begin{aligned} \prod_{v \in A} (|\langle x, v \rangle| + d(x, y)) &\geq \prod_{v \in A} 2^{-\frac{1}{2}} |\langle z, v \rangle| = 2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle g^{-1}y, v \rangle| = \\ &= 2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle y, gv \rangle| = 2^{-\frac{1}{2}|A|} \prod_{v \in gA} |\langle y, v \rangle| = 2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle y, v \rangle|. \end{aligned}$$

Thus,

$$\begin{aligned} w(B(x, d(x, y))) &\geq c_{R,k}^{(1)} d^N(x, y) \prod_{A \in R/\sim} \left( \prod_{v \in A} (d(x, y) + |\langle x, v \rangle|) \right)^{k|A|} \geq \\ &\geq c_{R,k}^{(1)} d^N(x, y) \prod_{A \in R/\sim} \left( 2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle y, v \rangle| \right)^{k|A|} = c_{R,k}^{(2)} d^N(x, y) w(y). \end{aligned}$$

On the other hand,

$$w(B(x, d(x, y))) \geq c_{R,k}^{(1)} d^N(x, y) w(x).$$

Multiplying the last two equations, we complete the proof.  $\square$

**Lemma 2.4.** *Let  $1 \leq j \leq N$ . If  $x_j \geq y_j$ , then*

$$K_{\text{Dunkl},j}(x, y) \geq c_{R,k} \frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

If  $x_j \leq y_j$ , then

$$K_{\text{Dunkl},j}(x, y) \leq c_{R,k} \frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

*Proof.* The lower bound is identical to that in [4, Theorem 1.2].  $\square$

**2.2. Schatten class.** [Zhijie will add this later]

**2.3. Nearly weakly orthonormal sequences.** Rochberg and Semmes [11] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [3, 7, 10, 1, 2, 9, 11, 13, 12]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [7, Section 13] for a complete proof). For our purposes, we adapt the extension to the Dunkl setting  $(\mathbb{R}^N, \|\cdot\|, w)$  rather than recalling the general definition of NWO sequences and the general extension to spaces of homogeneous type.

**Definition 2.5.** Let  $\mathcal{D}$  be the dyadic lattice in  $\mathbb{R}^N$ . A sequence of functions  $\{e_Q\}_{Q \in \mathcal{D}}$  is called *nearly weakly orthonormal (NWO, in short) sequence* if

- (i)  $\text{supp}(e_Q) \subset cQ$  for some positive constant  $c$  independent of  $Q$ .
- (ii)  $\|e_Q\|_{L_\infty(\mathbb{R}^N, w(x)dx)} \leq w(Q)^{-\frac{1}{2}}$ .

**Lemma 2.6.** Let  $1 < p < \infty$  and let  $\mathcal{D}$  be the dyadic lattice in  $\mathbb{R}^N$ . If  $T \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$ , then

$$\left\| \left\{ \langle T e_Q, f_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \right\}_{Q \in \mathcal{D}} \right\|_{l_p(\mathcal{D})} \leq c_{N,p,w,c_1,c_2} \|T\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}$$

for every NWO systems  $\{e_Q\}_{Q \in \mathcal{D}}$  and  $\{f_Q\}_{Q \in \mathcal{D}}$  (with respective constants  $c_1$  and  $c_2$ ).

*Proof.* reference to [11] is inappropriate. Rochberg and Semmes do not consider weighted setting. we absolutely must provide correct reference. This property can be found in [11, (1.10)] (see also [7, Section 7]).  $\square$

### 3. COMPARISON OF SOBOLEV NORMS

**Theorem 3.1.** For every  $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$ ,  $p > N$ , we have

$$\left( \int_{\mathbb{R}^N \times \mathbb{R}^N} \left( \frac{d(x,y)}{|x-y|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{d(x,y)^{2N}} \right)^{\frac{1}{p}} \leq c_{p,R} \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

**Lemma 3.2.** For every convex polyhedral cone  $\Gamma$ , we have

$$\int_{\Gamma} \frac{dx}{|x-y|^{2N}} \geq c_{\Gamma} \text{dist}(y, \Gamma)^{-N}, \quad y \notin \Gamma.$$

*Proof.* Find a point  $z \in \Gamma$  such that

$$|y - z| = \text{dist}(y, \Gamma).$$

We have  $z + \Gamma \subset \Gamma$ . Thus,

$$\int_{\Gamma} \frac{dx}{|x-y|^{2N}} \geq \int_{z+\Gamma} \frac{dx}{|x-y|^{2N}} = \int_{\Gamma} \frac{dx}{|x-(y-z)|^{2N}}.$$

Clearly,

$$|x - (y - z)| \leq |x| + |y - z|.$$

Thus,

$$\int_{\Gamma} \frac{dx}{|x - y|^{2N}} \geq \int_{\Gamma} \frac{dx}{(|x| + |y - z|)^{2N}} = |y - z|^{-N} \int_{\Gamma} \frac{dx}{(|x| + 1)^{2N}}.$$

□

**Lemma 3.3.** *If  $p \in (N, 2N)$ , then*

$$\sup_{a>0} \int_{\mathbb{R}^N} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x|)^p} < \infty.$$

*Proof. Step 1:* we claim that, for every  $a > 0$ ,

$$\begin{aligned} & \int_{\mathbb{R}^N} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x|)^p} \leq \\ & \leq 4^p \int_{\mathbb{R}^{N-1} \times (-a, a)} \frac{|x|^{p-2N} dx}{(1 + |x'| + a)^p} + 3^{2N-p} \int_{\mathbb{R}^N} \frac{|x|^{p-2N} dx}{(1 + |x|)^p}. \end{aligned}$$

We clearly have

$$\begin{aligned} & \int_{\mathbb{R}^N} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x|)^p} = \\ & = \int_{\mathbb{R}^{N-1} \times (\frac{a}{2}, \frac{3a}{2})} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x|)^p} + \int_{\mathbb{R}^{N-1} \times \mathbb{R} \setminus (\frac{a}{2}, \frac{3a}{2})} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x|)^p}. \end{aligned}$$

If  $x_N \notin (\frac{a}{2}, \frac{3a}{2})$ , then  $|x_N - a| \geq \frac{1}{3}|x_N|$ . Hence,  $|x - ae_N| \geq \frac{1}{3}|x|$ . Thus,

$$\begin{aligned} & \int_{\mathbb{R}^{N-1} \times \mathbb{R} \setminus (\frac{a}{2}, \frac{3a}{2})} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x|)^p} \leq \\ & \leq \int_{\mathbb{R}^{N-1} \times \mathbb{R} \setminus (\frac{a}{2}, \frac{3a}{2})} \frac{3^{2N-p} |x|^{p-2N} dx}{(1 + |x|)^p} \leq \int_{\mathbb{R}^N} \frac{3^{2N-p} |x|^{p-2N} dx}{(1 + |x|)^p}. \end{aligned}$$

If  $x_N \in (\frac{a}{2}, \frac{3a}{2})$ , then (here, we using the notation  $x' = (x_1, \dots, x_{N-1})$ )

$$1 + |x| \geq 1 + \max\{|x'|, \frac{1}{2}a\} \geq \frac{1}{4}(1 + |x'| + a).$$

Thus,

$$\begin{aligned} & \int_{\mathbb{R}^{N-1} \times (\frac{a}{2}, \frac{3a}{2})} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x|)^p} \leq 4^p \int_{\mathbb{R}^{N-1} \times (\frac{a}{2}, \frac{3a}{2})} \frac{|x - ae_N|^{p-2N} dx}{(1 + |x'| + a)^p} = \\ & = 4^p \int_{\mathbb{R}^{N-1} \times (-\frac{a}{2}, \frac{a}{2})} \frac{|x|^{p-2N} dx}{(1 + |x'| + a)^p} \leq 4^p \int_{\mathbb{R}^{N-1} \times (-a, a)} \frac{|x|^{p-2N} dx}{(1 + |x'| + a)^p}. \end{aligned}$$

Combining all preceding estimates, we infer the claim in Step 1.

**Step 2:** we claim that

$$\sup_{a>0} \int_{\mathbb{R}^{N-1} \times (-a, a)} \frac{|x|^{p-2N} dx}{(1 + |x'| + a)^p} < \infty.$$

Passing to polar coordinates in  $\mathbb{R}^{N-1}$ , we equivalently rewrite the claim as

$$\sup_{a>0} \int_{\mathbb{R}_+ \times (0, a)} \frac{(r^2 + x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(1 + r + a)^p} < \infty.$$

If  $0 < a < 1$ , then

$$\int_{\mathbb{R}_+ \times (0, a)} \frac{(r^2 + x_N^2)^{\frac{p}{2} - N} r^{N-2} dr dx_N}{(1 + r + a)^p} \leq \int_{\mathbb{R}_+ \times (0, 1)} \frac{(r^2 + x_N^2)^{\frac{p}{2} - N} r^{N-2} dr dx_N}{(1 + r)^p}.$$

If  $a \geq 1$ , then

$$\begin{aligned} \int_{\mathbb{R}_+ \times (0, a)} \frac{(r^2 + x_N^2)^{\frac{p}{2} - N} r^{N-2} dr dx_N}{(1 + r + a)^p} &\leq \int_{\mathbb{R}_+ \times (0, a)} \frac{(r^2 + x_N^2)^{\frac{p}{2} - N} r^{N-2} dr dx_N}{(r + a)^p} = \\ &= \int_{(a, \infty) \times (0, a)} \frac{(r^2 + x_N^2)^{\frac{p}{2} - N} r^{N-2} dr dx_N}{(r + a)^p} + \\ &+ \int_0^a \int_0^r \frac{(r^2 + x_N^2)^{\frac{p}{2} - N} r^{N-2} dx_N dr}{(r + a)^p} + \int_0^a \int_0^{x_N} \frac{(r^2 + x_N^2)^{\frac{p}{2} - N} r^{N-2} dr dx_N}{(r + a)^p} \leq \\ &\leq \int_{(a, \infty) \times (0, a)} \frac{r^{p-2N} r^{N-2} dr dx_N}{r^p} + \\ &+ \int_0^a \int_0^r \frac{r^{p-2N} r^{N-2} dx_N dr}{a^p} + \int_0^a \int_0^{x_N} \frac{x_N^{p-2N} r^{N-2} dr dx_N}{a^p} = \\ &= \int_a^\infty r^{-N-1} dr + a^{-p} \int_0^a r^{p-N-1} dr + \frac{1}{N-1} a^{-p} \int_0^a x_N^{p-N-1} dx_N = \\ &= \frac{1}{N} a^{-N} + \frac{1}{p-N} a^{-N} + \frac{1}{(N-1)(p-N)} a^{-N} \leq \frac{1}{N} + \frac{1}{p-N} + \frac{1}{(N-1)(p-N)}. \end{aligned}$$

This yields the claim in Step 2.

Combining Steps 1 and 2, we complete the proof of the lemma.  $\square$

**Lemma 3.4.** *If  $p \in (N, 2N)$ , then*

$$\int_{\mathbb{R}^{N-1} \times \mathbb{R}_+} \frac{|x - y|^{p-2N} dx}{|x + z|^p} \leq c_{p,N} z_N^{-N}, \quad y, z \in \mathbb{R}^{N-1} \times \mathbb{R}_+.$$

*Proof.* By dilation invariance, we may assume without loss of generality that  $z_N = 1$ . By translation invariance in  $\mathbb{R}^{N-1}$ , we may assume without loss of generality that  $z_1 = \dots = z_{N-1} = 0$ . We, therefore, have

$$|x + z| = |x + e_N| \geq \max\{1, |x|\} \geq \frac{1}{2}(1 + |x|), \quad x \in \mathbb{R}^{N-1} \times \mathbb{R}_+.$$

Thus,

$$\begin{aligned} \int_{\mathbb{R}^{N-1} \times \mathbb{R}_+} \frac{|x - y|^{p-2N} dx}{|x + z|^p} &\leq 2^p \int_{\mathbb{R}^{N-1} \times \mathbb{R}_+} \frac{|x - y|^{p-2N} dx}{(1 + |x|)^p} \leq \\ &= 2^p \int_{\mathbb{R}^N} \frac{|x - y|^{p-2N} dx}{(1 + |x|)^p} = 2^p \int_{\mathbb{R}^N} \frac{|x - |y|e_N|^{p-2N} dx}{(1 + |x|)^p}. \end{aligned}$$

The assertion follows from Lemma 3.3.  $\square$

We need the following elementary version of the Hahn-Banach theorem.

**Lemma 3.5.** *Let  $K \subset \mathbb{R}^N$  be a convex set and let  $y \notin K$ . Let  $z \in K$  be the point closest to  $y$ . Affine hyperplane  $P$  passing through  $z$  and orthogonal to  $y - z$  splits the space into two half-spaces, one of them containing  $y$ , the other one containing  $K$ .*

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<sup>1</sup>It is exactly at this point we are using the assumption  $p < 2N$ .

*Proof.* By translation invariance, we may assume without loss of generality that  $z = 0$ . By rotation and dilation invariance, we may assume without loss of generality that  $y = e_N$ . By assumption,  $\text{dist}(e_N, K) = 1$ . We need to show that  $K$  is contained in the (closed) lower half-plane.

Assume the contrary: let  $x \in K$  be such that  $x_N > 0$ . Since  $0 \in K$  and since  $K$  is convex, it follows that  $\epsilon x \in K$  for every  $\epsilon \in (0, 1)$ . Thus,

$$\text{dist}(e_N, K) \leq |e_N - \epsilon x|, \quad \epsilon \in (0, 1).$$

Since  $\epsilon \in (0, 1)$  is arbitrary, it follows that

$$\text{dist}(e_N, K) \leq \inf_{\epsilon \in (0, 1)} |e_N - \epsilon x| = \inf_{\epsilon \in (0, 1)} \left( \epsilon^2 \sum_{k=1}^{N-1} x_k^2 + (1 - \epsilon x_N)^2 \right)^{\frac{1}{2}}.$$

Since  $x_N > 0$ , it follows that infimum is strictly less than 1. Thus,  $\text{dist}(e_N, K) < 1$ . However, it is granted in the lemma that  $\text{dist}(e_N, K) = 1$ . This contradiction shows that such an  $x$  cannot exist. This completes the proof.  $\square$

**Lemma 3.6.** *If  $p \in (N, 2N)$ , then*

$$\int_{\Gamma} \frac{|x - y|^{p-2N} dx}{|x - gy|^p} \leq c_p \text{dist}(gy, \Gamma)^{-N}, \quad y \in \Gamma.$$

*Proof.* Find a point  $z \in \Gamma$  such that

$$|gy - z| = \text{dist}(gy, \Gamma).$$

By Lemma 3.5, affine hyperplane passing through  $z$  and orthogonal to  $gy - z$  splits the space into two half-spaces, one of them containing  $gy$ , the other one containing  $\Gamma$  (denote this one by  $H$ ).

We have  $y \in H$ ,  $\Gamma \subset H$  and  $gy \notin H$ . So,

$$\int_{\Gamma} \frac{|x - y|^{p-2N} dx}{|x - gy|^p} \leq \int_H \frac{|x - y|^{p-2N} dx}{|x - gy|^p}.$$

Using Lemma 3.4, we write

$$\int_H \frac{|x - y|^{p-2N} dx}{|x - gy|^p} \leq c_{p,N} \text{dist}(gy, H)^{-N}.$$

Since  $z \in \partial H$  and since  $gy - z$  is orthogonal to  $\partial H$ , it follows that  $\text{dist}(gy, H) = |gy - z| = \text{dist}(gy, \Gamma)$ .  $\square$

*Proof of Theorem 3.1. Case 1:* Let  $p \in (N, 2N)$ .

We have

$$\begin{aligned} & \left( \int_{\Gamma \times g\Gamma} \left( \frac{d(x, y)}{|x - y|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{d(x, y)^{2N}} \right)^{\frac{1}{p}} = \\ & = \left( \int_{\Gamma \times \Gamma} \left( \frac{|x - y|}{|x - gy|} \right)^p |f(x) - f(gy)|^p \frac{dxdy}{|x - y|^{2N}} \right)^{\frac{1}{p}} \leq \\ & \leq \left( \int_{\Gamma \times \Gamma} \left( \frac{|x - y|}{|x - gy|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{|x - y|^{2N}} \right)^{\frac{1}{p}} + \\ & + \left( \int_{\Gamma \times \Gamma} \left( \frac{|x - y|}{|x - gy|} \right)^p |f(y) - f(gy)|^p \frac{dxdy}{|x - y|^{2N}} \right)^{\frac{1}{p}} \leq \\ & \leq \left( \int_{\Gamma \times \Gamma} |f(x) - f(y)|^p \frac{dxdy}{|x - y|^{2N}} \right)^{\frac{1}{p}} + \end{aligned}$$



$$+ \left( \int_{\Gamma} |f(y) - f(gy)|^p \left( \int_{\Gamma} \frac{|x-y|^{p-2N}}{|x-gy|^p} dx \right) dy \right)^{\frac{1}{p}}.$$

Obviously, the first summand on the right hand side is less than  $\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}$ . By Lemmas 3.6 and 3.2,

$$\begin{aligned} & \left( \int_{\Gamma \times g\Gamma} \left( \frac{d(x,y)}{|x-y|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{d(x,y)^{2N}} \right)^{\frac{1}{p}} \leq \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)} + \\ & + c_{p,R}^{(1)} \left( \int_{\Gamma} |f(y) - f(gy)|^p \left( \int_{\Gamma} \frac{dx}{|x-gy|^{2N}} \right) dy \right)^{\frac{1}{p}} = \\ & = \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)} + c_{p,R}^{\frac{1}{p}} \left( \int_{\Gamma \times \Gamma} \frac{|f(y) - f(gy)|^p dxdy}{|x-gy|^{2N}} \right)^{\frac{1}{p}} \leq \\ & \leq \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)} + c_{p,R}^{(1)} \left( \int_{\Gamma \times \Gamma} \frac{|f(x) - f(gy)|^p dxdy}{|x-gy|^{2N}} \right)^{\frac{1}{p}} + \\ & + c_{p,R}^{(1)} \left( \int_{\Gamma \times \Gamma} \frac{|f(x) - f(y)|^p dxdy}{|x-gy|^{2N}} \right)^{\frac{1}{p}}. \end{aligned}$$

Now,

$$\left( \int_{\Gamma \times \Gamma} \frac{|f(x) - f(gy)|^p dxdy}{|x-gy|^{2N}} \right)^{\frac{1}{p}} = \left( \int_{\Gamma \times g\Gamma} \frac{|f(x) - f(y)|^p dxdy}{|x-y|^{2N}} \right)^{\frac{1}{p}} \leq \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}$$

and

$$\left( \int_{\Gamma \times \Gamma} \frac{|f(x) - f(y)|^p dxdy}{|x-gy|^{2N}} \right)^{\frac{1}{p}} \leq \left( \int_{\Gamma \times \Gamma} \frac{|f(x) - f(y)|^p dxdy}{|x-y|^{2N}} \right)^{\frac{1}{p}} \leq \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Thus,

$$\left( \int_{\Gamma \times g\Gamma} \left( \frac{d(x,y)}{|x-y|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{d(x,y)^{2N}} \right)^{\frac{1}{p}} \leq (1 + 2c_{p,R}^{(1)}) \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Summing over  $\Gamma$  and  $1 \neq g \in G$ , we obtain

$$\left( \int_{\mathbb{R}^N \times \mathbb{R}^N} \left( \frac{d(x,y)}{|x-y|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{d(x,y)^{2N}} \right)^{\frac{1}{p}} \leq |G|^2 (1 + 2c_{p,R}^{(1)}) \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

**Case 2:** Let  $p \geq 2N$ . We have

$$\begin{aligned} & \left( \int_{\Gamma \times g\Gamma} \left( \frac{d(x,y)}{|x-y|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{d(x,y)^{2N}} \right)^{\frac{1}{p}} \leq \\ & \leq \left( \int_{\Gamma \times g\Gamma} \left( \frac{d(x,y)}{|x-y|} \right)^{2N} |f(x) - f(y)|^p \frac{dxdy}{d(x,y)^{2N}} \right)^{\frac{1}{p}} = \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}. \end{aligned}$$

□

## 4. PROOF OF THEOREM 1.1 (i)

Let  $(X, \mu)$  be a measure space and let  $L_p(L_{q,\infty})^{\text{symm}}(X \times X, \mu \times \mu)$  denote the mixed-norm space consisting of all  $\mu \times \mu$ -measurable functions  $f : X \times X \rightarrow \mathbb{C}$  such that

$$\int_X \|f(x, \cdot)\|_{L_{q,\infty}(X,\mu)}^p d\mu(x), \int_X \|f(\cdot, y)\|_{L_{q,\infty}(X,\mu)}^p d\mu(y) < \infty.$$

We equip  $L_p(L_{q,\infty})^{\text{symm}}(X \times X, \mu \times \mu)$  with its natural norm.

**Lemma 4.1.** *Let  $(X, \mu)$  be a measure space and let  $2 < p < \infty$ . Suppose that  $K \in L_p(L_{p',\infty})^{\text{symm}}(X \times X, \mu \times \mu)$ . The integral operator*

$$T_K f(x) := \int_X K(x, y) f(y) d\mu(y)$$

*belongs to  $S_{p,\infty}(L_2(X, d\mu))$ . Moreover, we have*

$$\|T_K\|_{S_{p,\infty}(L_2(X, d\mu))} \leq c_p \|K\|_{L_p(L_{p',\infty})^{\text{symm}}(X \times X, \mu \times \mu)}.$$

*Proof.* The proof is given in [8, Lemma 2] (see also [7, Corollary 11.2]).  $\square$

**Lemma 4.2.** *Let  $p > 2$  and let  $\frac{1}{q} = 1 - \frac{2}{p}$ . We have*

$$\|F_1 F_2\|_{L_p(L_{p',\infty})(X \times X, \mu \times \mu)} \leq 2^{\frac{1}{pp'}} \|F_1\|_{L_p(X \times X, \mu \times \mu)} \|F_2\|_{L_\infty(L_{q,\infty})(X \times X, \mu \times \mu)}.$$

*Proof.* We clearly have

$$LHS = \left( \int_X \|F_1(x, \cdot) F_2(x, \cdot)\|_{L_{p',\infty}(X,\mu)}^p d\mu(x) \right)^{\frac{1}{p}}.$$

Clearly,  $\frac{1}{p'} = \frac{1}{p} + \frac{1}{q}$ . Thus,

$$\begin{aligned} \|F_1(x, \cdot) F_2(x, \cdot)\|_{L_{p',\infty}(X,\mu)} &\leq 2^{\frac{1}{p'}} \|F_1(x, \cdot)\|_{L_{p,\infty}(X,\mu)} \|F_2(x, \cdot)\|_{L_{q,\infty}(X,\mu)} \\ &\leq 2^{\frac{1}{p'}} \|F_1(x, \cdot)\|_{L_p(X,\mu)} \|F_2(x, \cdot)\|_{L_{q,\infty}(X,\mu)}. \end{aligned}$$

Hence,

$$\begin{aligned} LHS &\leq 2^{\frac{1}{pp'}} \left( \int_X \|F_1(x, \cdot)\|_{L_p(X,\mu)}^p \|F_2(x, \cdot)\|_{L_{q,\infty}(X,\mu)}^p d\mu(x) \right)^{\frac{1}{p}} \leq \\ &\leq 2^{\frac{1}{pp'}} \left( \int_X \|F_1(x, \cdot)\|_{L_p(X,\mu)}^p d\mu(x) \right)^{\frac{1}{p}} \times \sup_{x \in X} \|F_2(x, \cdot)\|_{L_{q,\infty}(X,\mu)} = \\ &= 2^{\frac{1}{pp'}} \|F_1\|_{L_p(X \times X, \mu \times \mu)} \|F_2\|_{L_\infty(L_{q,\infty})(X \times X, \mu \times \mu)}. \end{aligned}$$

$\square$

**Lemma 4.3.** *Let  $(X, \mu)$  be a measure space and let  $2 < p < \infty$ . Suppose that  $F \in L_p(X \times X, H^2(\mu \times \mu))$  and  $H \in L_\infty(L_{1,\infty})^{\text{symm}}(X \times X, \mu \times \mu)$ . We have*

$$\|T_{FH}\|_{S_{p,\infty}(L_2(X, d\mu))} \leq c_p \|F\|_{L_p(X \times X, H^2(\mu \times \mu))} \|H\|_{L_\infty(L_{1,\infty})^{\text{symm}}(X \times X, \mu \times \mu)}^{1-\frac{2}{p}}.$$

*Proof.* By Lemma 4.1,

$$\|T_{FH}\|_{S_{p,\infty}(L_2(X, d\mu))} \leq c_p \|FH\|_{L_p(L_{p',\infty})^{\text{symm}}(X \times X, \mu \times \mu)}.$$

Let  $\frac{1}{q} = 1 - \frac{2}{p}$ . It follows from Lemma 4.2 that

$$\|F_1 F_2\|_{L_p(L_{p',\infty})^{\text{symm}}(X \times X, \mu \times \mu)} \leq$$

$$\leq 2^{\frac{1}{pp'}} \|F_1\|_{L_p(X \times X, \mu \times \mu)} \|F_2\|_{L_\infty(L_{q,\infty})^{\text{symm}}(X \times X, \mu \times \mu)}.$$

Setting  $F_1 = FH^{\frac{2}{p}}$  and  $F_2 = H^{\frac{1}{q}}$ , we obtain

$$\begin{aligned} & \|FH\|_{L_p(L_{p',\infty})^{\text{symm}}(X \times X, \mu \times \mu)} \leq \\ & \leq 2^{\frac{1}{pp'}} \|FH^{\frac{2}{p}}\|_{L_p(X \times X, \mu \times \mu)} \|H^{\frac{1}{q}}\|_{L_\infty(L_{q,\infty})^{\text{symm}}(X \times X, \mu \times \mu)} = \\ & = 2^{\frac{1}{pp'}} \|F\|_{L_p(X \times X, H^2(\mu \times \mu))} \|H\|_{L_\infty(L_{1,\infty})^{\text{symm}}(X \times X, \mu \times \mu)}^{\frac{1}{q}} = \\ & = 2^{\frac{1}{pp'}} \|F\|_{L_p(X \times X, H^2(\mu \times \mu))} \|H\|_{L_\infty(L_{1,\infty})^{\text{symm}}(X \times X, \mu \times \mu)}^{1-\frac{2}{p}}. \end{aligned}$$

□

**Lemma 4.4.** *Let  $(X, \mu)$  be a measure space and let  $2 < p < \infty$ . Suppose that  $F \in L_p(X \times X, GH^2(\mu \times \mu))$  and  $H \in L_\infty(L_{1,\infty})^{\text{symm}}(X \times X, \mu \times \mu)$ . We have*

$$\|T_{FH}\|_{S_p(L_2(X, d\mu))} \leq c_{p,H} \|F\|_{L_p(X \times X, H^2(\mu \times \mu))}.$$

*Proof.* The assertion follows from Lemma 4.3 by real interpolation. □

The following proposition, strictly speaking, is not part of the proof of Theorem 1.1 (i). Instead, it features in the proof of Theorem 1.1 (ii) in the next section. However, in terms of the employed technique, it certainly belongs to this section.

**Proposition 4.5.** *Let  $p > 2$ , let  $\Gamma$  be a chamber and let  $f \in \dot{W}^{\frac{N}{p},p}(\Gamma)$ . Let  $V$  be an integral operator whose integral kernel  $K$  satisfies the condition*

$$|K(x, y)| \leq \frac{1}{w(B(x, |x - y|))}.$$

*For every  $1 \leq j \leq N$ , we have*

$$\|[V, M_f]\|_{\mathcal{L}_p(L_2(\Gamma, w(x)dx))} \leq c_{R,k} \|f\|_{\dot{W}^{\frac{N}{p},p}(\Gamma)}.$$

*Proof.* In what follows, let  $X = \Gamma$  and  $d\mu(x) = w(x)dx$ . Set

$$F(x, y) = w(B(x, |x - y|))K(x, y)(f(x) - f(y)), \quad x \neq y,$$

$$H(x, y) = \frac{1}{w(B(x, |x - y|))}, \quad x \neq y.$$

By [7, Lemma 5.7],  $H \in L_\infty(L_{1,\infty})^{\text{symm}}(\Gamma \times \Gamma, \mu \times \mu)$ . By assumption on the kernel  $K$ ,

$$\|F\|_{L_p(\Gamma \times \Gamma, G^2(\mu \times \mu))} \leq \left( \int_{\Gamma \times \Gamma} |f(x) - f(y)|^p \frac{w(x)w(y)dxdy}{w^2(B(x, |x - y|))} \right)^{\frac{1}{p}}.$$

By (2.1),

$$w(B(x, |x - y|)) \geq c'_{R,k} |x - y|^N w^{\frac{1}{2}}(x) w^{\frac{1}{2}}(y), \quad x, y \in \mathbb{R}^N.$$

Thus,

$$\begin{aligned} & \|F\|_{L_p(\Gamma \times \Gamma, H^2(\mu \times \mu))} \leq \\ & \leq (c'_{R,k})^{-\frac{2}{p}} \left( \int_{\Gamma \times \Gamma} \frac{|f(x) - f(y)|^p}{|x - y|^{2N}} dxdy \right)^{\frac{1}{p}} = (c'_{R,k})^{-\frac{2}{p}} \|f\|_{\dot{W}^{\frac{N}{p},p}(\Gamma)}. \end{aligned}$$

Obviously, the integral kernel of the operator  $[V, M_f]$  is  $FH$ . The assertion follows now from Lemma 4.4. □

*Proof of Theorem 1.1 (i).* In what follows, let  $X = \mathbb{R}^N$  and  $d\mu(x) = w(x)dx$ . Set

$$F(x, y) = w(B(x, d(x, y)))K_{\text{Dunkl},j}(x, y)(f(x) - f(y)), \quad x \neq y,$$

$$H(x, y) = \frac{1}{w(B(x, d(x, y)))}, \quad x \neq y.$$

By [7, Lemma 11.7],  $H \in L_\infty(L_{1,\infty})^{\text{symm}}(X \times X, \mu \times \mu)$ . By Lemma 2.2,

$$\begin{aligned} & \|F\|_{L_p(X \times X, H^2(\mu \times \mu))} \leq \\ & \leq c_{R,k}^{(1)} \left( \int_{\mathbb{R}^N \times \mathbb{R}^N} \left( \frac{d(x, y)}{|x - y|} \right)^p |f(x) - f(y)|^p \frac{w(x)w(y)dxdy}{w(B(x, d(x, y)))^2} \right)^{\frac{1}{p}}. \end{aligned}$$

Using Lemma 2.3, we conclude that

$$\|F\|_{L_p(X \times X, H^2(\mu \times \mu))} \leq c_{R,k}^{(2)} \left( \int_{\mathbb{R}^N \times \mathbb{R}^N} \left( \frac{d(x, y)}{|x - y|} \right)^p |f(x) - f(y)|^p \frac{dxdy}{d(x, y)^{2N}} \right)^{\frac{1}{p}}.$$

Using Theorem 3.1, we conclude that

$$\|F\|_{L_p(X \times X, H^2(\mu \times \mu))} \leq c_{R,k} \|f\|_{\dot{W}^{\frac{N}{p}, p}(\mathbb{R}^N)}.$$

Obviously, the integral kernel of the operator  $[R_{\text{Dunkl},j}, M_f]$  is  $FH$ . The assertion follows now from Lemma 4.4.  $\square$

## 5. PROOF OF THEOREM 1.1 (ii)

**Lemma 5.1.** *Let  $Q \in \mathcal{D}$ . We have*

$$K_j(x, y) \geq \frac{c_k}{w(Q)}, \quad x \in 3Q, \quad y \in Q + 3s(Q)e_j.$$

*Proof.* TO BE DONE. Follows from Lemma 2.4.  $\square$

**Lemma 5.2.** *Let real-valued  $f \in L_1^{\text{loc}}(\mathbb{R}^N)$  be such that the commutator  $[R_{\text{Dunkl},j}, M_f]$  is bounded on  $L_2(\mathbb{R}^N, w(x)dx)$ . For every  $Q \in \mathcal{D}$ , there exist  $\xi_Q$  and  $\eta_Q$  supported in  $7Q$  such that  $\|\xi_Q\|_\infty, \|\eta_Q\|_\infty \leq m(Q)^{-\frac{1}{2}}$  and such that*

$$\frac{1}{w(3Q)} \int_{3Q} |f(y) - f_3Q| w(y) dy \leq c_k |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

*Proof.* Let  $\alpha$  be the median of  $f$  on  $3Q$  and  $\beta$  be the median of  $f$  on  $Q + 3s(Q)e_j$ . Assume for definiteness  $\alpha \geq \beta$ . Set

$$A_1 = \{x \in 3Q : f(x) \geq \alpha\}, \quad B_1 = \{x \in Q + 3s(Q)e_j : f(x) \leq \beta\},$$

$$A_2 = \{x \in 3Q : f(x) \leq \beta\}, \quad B_2 = \{x \in Q + 3s(Q)e_j : f(x) \geq \alpha\},$$

$$\xi_{Q,s} = w(Q)^{-\frac{1}{2}} \chi_{A_s}, \quad \eta_{Q,s} = w(Q)^{-\frac{1}{2}} \chi_{B_s}, \quad s = 1, 2.$$

We have

$$\begin{aligned} & \langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} = \\ & = \frac{1}{w(Q)} \int_{B_s \times A_s} K_j(x, y) (f(y) - f(x)) w(x) w(y) dx dy. \end{aligned}$$

If  $x \in B_1$  and  $y \in A_1$ , then

$$\begin{aligned} & f(y) - f(x) = (f(y) - \alpha) + (\alpha - \beta) + (\beta - f(x)) = \\ & = |f(y) - \alpha| + |\alpha - \beta| + |f(x) - \beta| \geq |f(y) - \alpha| + |\alpha - \beta| \geq 0. \end{aligned}$$

If  $x \in B_2$  and  $y \in A_2$ , then

$$\begin{aligned} f(y) - f(x) &= (f(y) - \beta) + (\beta - f(x)) = \\ &= -|f(y) - \beta| - |f(x) - \beta| \leq -|f(y) - \beta| \leq 0. \end{aligned}$$

Thus,

$$\begin{aligned} &\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq \\ &\geq \frac{1}{w(Q)} \int_{B_1 \times A_1} K_j(x, y) (|f(y) - \alpha| + |\alpha - \beta|) w(x) w(y) dx dy \geq \\ &\geq \frac{c_k}{w^2(Q)} \int_{B_1 \times A_1} (|f(y) - \alpha| + |\alpha - \beta|) w(x) w(y) dx dy = \\ &= \frac{c_k}{w^2(Q)} \cdot \left( w(B_1) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(B_1) w(A_1) \right) \geq \\ &= \frac{c_k}{w^2(Q)} \cdot \left( \frac{1}{2} w(Q) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| \cdot \frac{1}{4} w^2(Q) \right) \geq \\ &\geq \frac{c_k}{4w(Q)} \cdot \left( \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(Q) \right) \end{aligned}$$

and

$$\begin{aligned} &-\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq \\ &\geq \frac{1}{w(Q)} \int_{B_2 \times A_2} K_j(x, y) |f(y) - \beta| w(x) w(y) dx dy \geq \\ &\geq \frac{c_k}{w^2(Q)} \int_{B_2 \times A_2} |f(y) - \beta| w(x) w(y) dx dy = \\ &= \frac{c_k}{w^2(Q)} \cdot w(B_2) \cdot \int_{A_2} |f(y) - \beta| w(y) dy \geq \\ &= \frac{c_k}{2w(Q)} \int_{A_2} |f(y) - \beta| w(y) dy. \end{aligned}$$

It remains to note that

$$\begin{aligned} &\int_{3Q} |f(y) - \alpha| w(y) dy = \int_{A_1} |f(y) - \alpha| w(y) dy + \\ &+ \int_{A_2} |f(y) - \alpha| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |f(y) - \alpha| w(y) dy \leq \\ &\leq \int_{A_1} |f(y) - \alpha| w(y) dy + \int_{A_2} |f(y) - \beta| w(y) dy \\ &+ \int_{A_2} |\alpha - \beta| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |\alpha - \beta| w(y) dy \leq \\ &\leq \int_{A_1} |f(y) - \alpha| w(y) dy + \int_{A_2} |f(y) - \beta| w(y) dy + 2|\alpha - \beta| w(3Q). \end{aligned}$$

Combining the last 3 equations, we conclude

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - \alpha| w(y) dy \leq \max_{s=1,2} |\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

If

$$|\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}| \geq |\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|,$$

then we set  $\xi_Q = \xi_{Q,1}$  and  $\eta = \eta_{Q,1}$ . Otherwise, set  $\xi_Q = \xi_{Q,2}$  and  $\eta_Q = \eta_{Q,2}$ . Thus,

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - \alpha| w(y) dy \leq |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x) dx)}|.$$

Hence,

$$c'_k |f_{3Q} - \alpha| \leq |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x) dx)}|$$

and, finally,

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - f_{3Q}| w(y) dy \leq 2 |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x) dx)}|.$$

□

**Lemma 5.3.** *Suppose that  $1 < p < \infty$  and  $f \in L_1^{\text{loc}}(\mathbb{R}^N)$ . If  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x) dx))$  for some  $j \in \{1, 2, \dots, N\}$ , then*

$$\begin{aligned} & \left\| \left\{ \frac{1}{w(3Q)} \int_{3Q} |f(x) - f_{3Q}| w(x) dx \right\}_{Q \in \mathcal{D}} \right\|_{l_p(\mathcal{D})} \leq \\ & \leq c_k \| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x) dx))}. \end{aligned}$$

*Proof.* Let  $\{\xi_Q\}_{Q \in \mathcal{D}}$  and  $\{\eta_Q\}_{Q \in \mathcal{D}}$  be functions constructed in Lemma 5.2. By Lemma 2.6,

$$\left\{ \langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x) dx)} \right\}_{Q \in \mathcal{D}} \in l_p.$$

The assertion follows now from Lemma 5.2. □

**Lemma 5.4.** *Let  $Q$  be a cube in  $\mathbb{R}^N$  (with sides parallel to coordinate hyperplanes). We have*

$$\sup_{x \in Q} w(x) \leq \frac{c_k}{m(Q)} \int_Q w(x) dx.$$

*Proof.* If  $x = c_Q + y$ ,  $y \in [-\frac{1}{2}s(Q), \frac{1}{2}s(Q)]^d$ , then

$$|\langle x, v \rangle| \leq |\langle c_Q, v \rangle| + r |\langle y, v \rangle| \leq |\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q).$$

Thus,

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{k(v)} \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q))^{k(v)}.$$

In particular,

$$LHS \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q))^{k(v)}.$$

Fix chamber  $\Gamma$  such that  $c_Q \in \Gamma$ . We have

$$\int_Q w(x) dx \geq \int_{c_Q + (B(0, \frac{1}{2}s(Q)) \cap \Gamma)} w(x) dx = \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma} w(c_Q + y) dy.$$

If  $y \in \Gamma$ , then

$$|\langle c_Q + y, v \rangle| = |\langle c_Q, v \rangle| + |\langle y, v \rangle|, \quad v \in R.$$

Thus,

$$w(c_Q + y) = \prod_{v \in R} (|\langle c_Q, v \rangle| + |\langle y, v \rangle|)^{k(v)}.$$

Hence,

$$\int_Q w(x)dx \geq \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma} \prod_{v \in R} (|\langle c_Q, v \rangle| + |\langle y, v \rangle|)^{k(v)} dy.$$

Fix a closed cone  $\Gamma' \subset \Gamma$  such that (i) interior of  $\Gamma'$  is non-empty and (ii)  $\partial\Gamma \cap \partial\Gamma' = \{0\}$ . We have

$$|\langle y, v \rangle| \geq c_{\Gamma'} |y|, \quad y \in \Gamma', \quad v \in R.$$

We, therefore, have

$$\begin{aligned} \int_Q w(x)dx &\geq \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma'} \prod_{v \in R} (|\langle c_Q, v \rangle| + c_{\Gamma'} |y|)^{k(v)} dy \geq \\ &\geq \int_{\substack{y \in \Gamma' \\ \frac{1}{4}s(Q) \leq |y| \leq \frac{1}{4}s(Q)}} \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{1}{2}c_{\Gamma'} s(Q))^{k(v)} dy = \\ &= s(Q)^N \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{1}{4}c_{\Gamma'} s(Q))^{k(v)} \times (2^{-N} - 4^{-N}) \int_{\substack{y \in \Gamma' \\ |y| \leq 1}} dy. \end{aligned}$$

Combining this with the preceding paragraph, we complete the proof.  $\square$

**Lemma 5.5.** *Suppose  $1 < t < 1 + \frac{1}{|R| \cdot \max_{v \in R} k(v)}$ . Let  $Q$  be a cube in  $\mathbb{R}^N$  (with sides parallel to coordinate hyperplanes). We have*

$$\left( \frac{\int_Q w^{1-t}(x)dx}{\int_Q w(x)dx} \right)^{\frac{1}{t}} \leq c_{t,k} \frac{m(Q)}{\int_Q w(x)dx}.$$

*Proof.* If  $v \in R$  is such that

$$\sqrt{d}s(Q) \leq |\langle c_Q, v \rangle|,$$

then

$$|\langle x, v \rangle| \geq |\langle c_Q, v \rangle| - |\langle x - c_Q, v \rangle| \geq |\langle c_Q, v \rangle| - \frac{\sqrt{d}}{2}s(Q) \geq \frac{1}{2}|\langle c_Q, v \rangle|, \quad x \in Q.$$

In this case,

$$\begin{aligned} \int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx &\leq 2^{(t-1)|R|k(v)} |\langle c_Q, v \rangle|^{(1-t)|R|k(v)} \cdot \int_Q dx = \\ &= 2^{(t-1)|R|k(v)} |\langle x_0, v \rangle|^{(1-t)|R|k(v)} \cdot s(Q)^N. \end{aligned}$$

If  $v \in R$  is such that

$$|\langle c_Q, v \rangle| \leq \sqrt{d}s(Q),$$

then let  $y_0 = \langle c_Q, v \rangle$  and  $z_0 = c_Q - y_0 v$ . Let us introduce new variables  $y = \langle x, v \rangle$  and  $z = x - yv$  (it varies in the plane orthogonal to  $v$ ). We write

$$\begin{aligned} \int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx &\leq \int_{|x - c_Q| \leq \frac{\sqrt{d}}{2}s(Q)} |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \leq \\ &\leq \int_{|y - y_0|, |z - z_0| \leq \frac{\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)} dy dz = \\ &= \left( \frac{\sqrt{d}}{2}s(Q) \right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y - y_0| \leq \frac{\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)} dy \leq \\ &= \left( \frac{\sqrt{d}}{2}s(Q) \right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y| \leq \frac{3\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)} dy = \end{aligned}$$

$$= s(Q)^{N+(1-t)|R|k(v)} \cdot \left(\frac{\sqrt{d}}{2}\right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y| \leq \frac{3\sqrt{d}}{2}} |y|^{(1-t)|R|k(v)} dy.$$

Combining the estimates in the preceding paragraphs, we write

$$\int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \leq c_1 s(Q)^N (\max\{|\langle c_Q, v \rangle|, r\})^{(1-t)|R|k(v)}.$$

By Hölder inequality,

$$\begin{aligned} \int_Q w^{1-t}(x) dx &\leq \prod_{v \in R} \left( \int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \right)^{\frac{1}{|R|}} \leq \\ &\leq c_2 s(Q)^N \prod_{v \in R} (\max\{|\langle c_Q, v \rangle|, s(Q)\})^{(1-t)k(v)}. \end{aligned}$$

The assertion follows easily.  $\square$

**Lemma 5.6.** *Measures  $dx$  and  $w(x)dx$  (and distance  $(x, y) \rightarrow |x - y|_\infty$ ) satisfy the conditions in Proposition 1.2 in [6].*

*Proof.* That both measures are doubling is obvious.

Let  $Q$  be a cube in  $\mathbb{R}^N$  (with sides parallel to coordinate hyperplanes). By Lemmas 5.4 and 5.5,

$$\begin{aligned} \left( \frac{1}{m(Q)} \int_Q w^t(x) dx \right)^{\frac{1}{t}} &\leq \frac{c_k}{m(Q)} \int_\Gamma w(x) dx, \quad t > 0. \\ \left( \frac{\int_Q w^{1-t}(x) dx}{\int_Q w(x) dx} \right)^{\frac{1}{t}} &\leq c_G \frac{m(Q)}{\int_Q w(x) dx}, \quad 1 < t < 1 + \frac{1}{|R| \cdot \max_{v \in R} k(v)}. \end{aligned}$$

The assertion follows now from Proposition 3.6 in [6].  $\square$

*Proof of Theorem 1.1 (ii).* The left hand side in Lemma 5.3 is the oscillation semi-norm of the function  $f$  with respect to measure  $w(x)dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$ . So, the oscillation semi-norm of the function  $f$  with respect to measure  $w(x)dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$  is controlled by the  $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator  $[R_{\text{Dunkl}, j}, M_f]$ .

Lemma 5.6 asserts that measures  $dx$  and  $w(x)dx$  (and distance  $(x, y) \rightarrow |x - y|_\infty$ ) satisfy the conditions in Proposition 1.2 in [6]. Appealing to Proposition 1.2 in [6], we conclude that the oscillation semi-norm of the function  $f$  with respect to measure  $dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$  is controlled by the one with respect to measure  $w(x)dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$ . By the preceding paragraph, the oscillation semi-norm of the function  $f$  with respect to measure  $dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$  is controlled by the  $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator  $[R_{\text{Dunkl}, j}, M_f]$ .

That oscillation semi-norm of the function  $f$  with respect to measure  $dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$  is equivalent to the Sobolev semi-norm of the function  $f$  is a standard fact. The proof (in much more general setting) is available in Proposition 1.29 in [7]. The assertion follows now from the preceding paragraph.  $\square$



## 6. CONSTANT CHARACTERIZATION

**Theorem 6.1.** *If  $f \in L_1^{\text{loc}}(\mathbb{R}^N, w(x)dx)$  is such that*

$$[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$$

*for some  $p \leq d$ , then  $f$  is constant.*

*Proof.* By Lemma 5.6, measures  $dx$  and  $w(x)dx$  and operator  $T_{\Gamma,j} = M_{\chi_\Gamma} R_{\text{Dunkl},j} M_{\chi_\Gamma}$  satisfy on  $\Gamma$  the assumptions in Theorem 1.4 in [6]. By assumption,  $[T_{\Gamma,j}, M_f] \in \mathcal{L}_p(L_2(\Gamma, w(x)dx))$ . By Theorem 1.4 in [6],  $f$  is constant on  $\Gamma$ .

Hence,  $f$  is constant on every chamber. Let  $f = c_1$  on  $\Gamma_1$  and  $f = c_2$  on  $\Gamma_2$ . On the other hand,  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{2N}(L_2(\mathbb{R}^N, w(x)dx))$ . By Theorem 1.1 (ii),  $f \in \dot{W}^{\frac{1}{2}, 2N}(\mathbb{R}^N)$ . We have

$$\begin{aligned} \int_{\Gamma_1 \times \Gamma_2} \frac{|c_1 - c_2|^{2N} dx dy}{|x - y|^{2N}} &= \int_{\Gamma_1 \times \Gamma_2} \frac{|f(x) - f(y)|^{2N} dx dy}{|x - y|^{2N}} \leq \\ &\leq \int_{\mathbb{R}^N \times \mathbb{R}^N} \frac{|f(x) - f(y)|^{2N} dx dy}{|x - y|^{2N}} = \|f\|_{\dot{W}^{\frac{1}{2}, 2N}(\mathbb{R}^N)}^{2N} < \infty. \end{aligned}$$

It follows that  $c_1 = c_2$ . Since chambers  $\Gamma_1$  and  $\Gamma_2$  are arbitrary, the assertion follows.  $\square$

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